

COMMENTS AND RESPONSES

Disclosing Pleiotropic Effects During Genetic Risk Assessment for Alzheimer Disease

TO THE EDITOR: Nobel laureate Dr. James Watson, recognized for his contribution to the discovery of DNA structure, was among the first persons to undergo whole-genome sequencing (1). He requested to receive all data generated from this research project except the disclosure of his apolipoprotein E (*APOE*) genotype (1).

Christensen and colleagues (2) show that disclosing pleiotropic effects of the *APOE* genotype does not increase anxiety or depression among cognitively healthy adults. In fact, the disclosure of risk for Alzheimer disease was associated with improved healthy behaviors when combined with information about the influence of *APOE* on coronary artery disease. However, physicians can recommend behaviors to improve cardiovascular health and patients can decide to follow this advice even if genetic information on risk for Alzheimer disease is not disclosed.

Although Christensen and colleagues (2) replicate previous findings of the REVEAL (Risk Evaluation and Education for Alzheimer's Disease) study group on short-term psychological outcomes of *APOE* genetic risk disclosure (3), the assessment of anxiety and depression in healthy persons cannot predict how those who know that they carry 1 or 2 *APOE* ϵ 4 risk alleles will feel when they experience changes in cognitive performance in the future, regardless of whether such changes are related to normal aging or a neurodegenerative disease. The exact mechanisms by which *APOE* allelic variants mediate disease risk remain largely unknown. The *APOE* ϵ 4 risk allele could even have antagonistically pleiotropic effects, which means that *APOE* ϵ 4 carrier status may be associated with risks and benefits across different stages of life (4).

Furthermore, it has recently been shown that *APOE* genotype disclosure may have substantial adverse effects on subjective and objective memory performance in cognitively healthy older adults (5). Knowledge of *APOE* status could distort clinical assessments with a bias toward false-positive diagnoses of cognitive impairment among *APOE* ϵ 4 carriers or could interfere with patient homogeneity in clinical trials on Alzheimer disease (5). Therefore, we believe that the potential benefits caused by acknowledgment of *APOE* pleiotropy do not outweigh the challenges associated with the disclosure of genetic risk for Alzheimer disease.

Robert Haussmann, MD

Markus Donix, MD

University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany

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Appropriate Antibiotic Use for Acute Respiratory Tract Infection in Adults

TO THE EDITOR: Harris and colleagues' advice that "clinicians should test patients with symptoms suggestive of group A streptococcal pharyngitis . . . by rapid antigen detection test and/or culture for group A *Streptococcus*" (1) does not agree with European guidelines (2). The problem with group A streptococcal pharyngitis is that clinical findings and tests, even in combination, are not sufficiently reliable to diagnose active infection. According to the guidelines from the United Kingdom's National Institute for Health and Care Excellence and the Scottish Intercollegiate Guidelines Network (3), neither rapid antigen detection testing nor throat swab culture can differentiate between the streptococcal carrier state and invasive infection.

Reflexive culture with throat swab has poor sensitivity (73% to 80%), and results take up to 48 hours to be reported—too late to substantially influence the clinical course (3). Asymptomatic carriage ranges from 6% to 40%: *Streptococcus* can live in our noses, mouths, and throats and not make us sick, which means that a positive result on rapid antigen detection testing or culture does not necessarily indicate that infection is present. Swabs may be useful in high-risk groups to guide the choice of treatment if the initial treatment fails (3).

Increasingly more European guidelines consider acute sore throat a self-limiting disease (2). Antibiotics are prescribed immediately only when the patient has risk factors, decreased immunocompetence, or extreme illness (approximately 5% of cases) (2). Antibiotics should not be used to secure symptomatic relief in sore throat (3). One or 2 days of extra doses of acetaminophen (paracetamol) (or nonsteroidal anti-inflammatory drugs) every 4 hours can replace unnecessary antibiotic use in most cases and might reduce the problem of antibiotic resistance. If the patient does not improve within 2 or 3 days, the physical examination must be repeated to exclude peritonsillar abscess and even Lemierre syndrome (4). More and more guidelines have concluded that, in

patients with respiratory infections in whom clinicians believe that it is safe not to prescribe antibiotics immediately, no antibiotics with advice to return if symptoms do not resolve is likely to result in the least antibiotic use while maintaining similar patient satisfaction (5).

Finally, we applaud that Harris and colleagues' guideline does not recommend tonsillectomy solely to reduce the frequency of group A streptococcal pharyngitis in adults (1).

Jan Matthys, MD
University of Ghent
Ghent, Belgium

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IN RESPONSE: We appreciate Dr. Matthys's comments and acknowledge that—even when patients with pharyngitis meet recommended criteria for group A streptococcal testing—as many as 20% of children and fewer adults will have a positive result from colonization rather than true disease (1). This factor leads to overuse of antibiotics for pharyngitis.

In a recent study examining outpatient antibiotic prescribing, pharyngitis was 1 of the conditions for which antibiotics were most commonly prescribed; 56% and 72% of pediatric and adult pharyngitis cases, respectively, resulted in an antibiotic prescription far exceeding estimates of the true prevalence of group A streptococcal infection and colonization (37% for children and 18% for adults) (2). This finding suggests that, in many cases, providers are either not testing for group A streptococcal infection or are prescribing antibiotics despite a negative test result. The 2012 Infectious Diseases Society of America clinical guideline for diagnosis and management of group A streptococcal pharyngitis recommends testing for group A streptococcal infection in patients with clinical symptoms suggestive of this condition and prescribing antibiotics for patients with a positive result for group A *Streptococcus* (3). An important first step in reducing unnecessary antibiotic use in the United States would be to

follow the Infectious Diseases Society of America's recommendation to treat only patients with positive results for group A *Streptococcus*.

We agree with Dr. Matthys that providers should offer recommendations for symptomatic relief. The Centers for Disease Control and Prevention has developed a symptomatic prescription pad that providers can use to guide symptom management in their patients (www.cdc.gov/getsmart/community/materials-references/print-materials/hcp/viral-rx-pad-color.pdf).

Although acute pharyngitis in adults is usually self-limited, antibiotic treatment in patients with confirmed group A streptococcal pharyngitis has been shown to provide modest benefit (3, 4). The goals of treatment in patients with this condition are to prevent nonsuppurative complications, reduce the duration of symptoms, and decrease transmission of group A *Streptococcus* to close contacts (3, 4). The magnitude of the benefit depends on the epidemiology of group A streptococcal infections in the region and the risk for poststreptococcal complications. In light of the growing threat of antibiotic resistance and increasing recognition of adverse events associated with antibiotic use, including community-associated *Clostridium difficile* infection, studies that carefully weigh the benefits and risks of antibiotic treatment for group A streptococcal pharyngitis are needed.

Aaron M. Harris, MD, MPH
Lauri A. Hicks, DO
Centers for Disease Control and Prevention
Atlanta, Georgia

Amir Qaseem, MD, PhD, MHA
American College of Physicians
Philadelphia, Pennsylvania

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Judging the Past: How History Should Inform Bioethics

TO THE EDITOR: I believe that the otherwise cogent reasoning in Lerner and Caplan's commentary on how history should

inform bioethics (1) suffers from 1 logical flaw, namely, ignoring the distinction between research and clinical care. All but 1 of the past ethical violations that they discuss are clearly research projects—that is, activities designed to yield generalizable knowledge. However, 1 example that they describe, “lobotomy to treat mental illness,” is not research but rather clinical care designed to benefit the individual patient. Although hindsight may show that this treatment usually was not beneficial and was perhaps often harmful, the physicians administering it believed that they were providing a beneficial treatment to the individual patient. The history of medicine is replete with treatments widely used without adequate scientific evidence of efficacy (as now understood) and that were later shown to be ineffective or harmful (for example, blood-letting, use of leeches, and prolonged bed rest after myocardial infarction). These may be examples of poor medical care, but they are conceptually distinct from violations of research ethics.

I also believe that when a quote by the fictional character Martin Arrowsmith is credited to the 1931 film *Arrowsmith*, the 1925 Sinclair Lewis novel of the same name on which the film was based should be mentioned, as well.

David A. Gorelick, MD, PhD
University of Maryland School of Medicine
Baltimore, Maryland

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Reference

1. Lerner BH, Caplan AL. Judging the past: how history should inform bioethics. *Ann Intern Med.* 2016;164:553-7. [PMID: 27089070] doi:10.7326/M15-2642

TO THE EDITOR: I read Lerner and Caplan's article on the importance of a less moralistic historical approach in analyzing some past bioethical scandals (1) with interest. I particularly appreciate the examples that they provided and agree with their conclusions that “learning how societal values, scientific zeal, ideological beliefs, and the desire for personal achievement influenced [researchers who made dubious ethical choices] reveals how similar factors can and often still remain in play, even in our supposedly more “enlightened” era (1). History has many things to teach medical students and physicians, even though Whig historians in past decades have sometimes distorted and misjudged the development of modern medicine in certain historical periods. For example, the Middle Ages are still viewed as a period of obscurantism in the history of medicine in contrast with the Enlightenment and positivism. I believe that history may support bioethicists in understanding that some modern bioethical issues were posed before the positivist thought of Claude Bernard and can be found from the Hippocratic oath onward.

In Italian medical schools, education and doctoral programs are often the same for medical historians and bioethicists. Furthermore, researchers and professors in these fields

belong to the same scientific disciplinary sector in the Italian academic system. I believe that the Italian experience provides a good model that may be exported into the academic curricula of other countries, thus allowing a better integration between history and bioethics as Lerner and Caplan propose. In this way, both disciplines would benefit.

Michele A. Riva, MD, PhD
University of Milano Bicocca
Monza, Italy

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1. Lerner BH, Caplan AL. Judging the past: how history should inform bioethics. *Ann Intern Med.* 2016;164:553-7. [PMID: 27089070] doi:10.7326/M15-2642

IN RESPONSE: Dr. Gorelick is correct that our reference to lobotomy is an example of a therapy as opposed to an experiment. We included lobotomy because it is such a good example of an intervention seen as ethically dubious today that made sense in the past. In this case, physicians in the post-World War II era embraced it because treatment options for psychotic patients were few and ongoing institutionalization was undesirable. Indeed, more historical analysis is warranted of the examples that Dr. Gorelick raises, such as why past physicians (and patients) embraced certain treatments as effective when later research showed them to be useless or even harmful. We also thank Dr. Gorelick for mentioning Sinclair Lewis' seminal novel, which should ideally be read by all medical students.

Dr. Riva is correct that teaching history and bioethics concurrently is advantageous because it enhances the understanding of both fields. In the United States, medical schools generally teach these topics—and related subjects, such as literature and medicine—in separate lecture series.

Barron H. Lerner, MD, PhD
Arthur L. Caplan, PhD
New York University School of Medicine
New York, New York

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The Deceptive Appeal of Direct-to-Consumer Genetics

TO THE EDITOR: I read Burke and Trinidad's editorial (1) with interest. Direct-to-consumer (DTC) genetic testing is appealing despite policy statements from professional organizations warning that it should never be done without input from a primary care provider (PCP) and, ideally, a credentialed genetics professional (CGP) (2). This testing offers the consumer

personal autonomy, but consumers must be educated that it may be grossly incomplete or inappropriate (3).

During the past year, 5 patients have been referred to my practice for evaluation of results of DTC genetic testing that revealed that they had 1 of the *BRCA* founder mutations in persons of Ashkenazi Jewish ancestry; they were totally unprepared for and devastated by the results. In each case, the PCP correctly recognized that further evaluation was appropriate. This testing is unfortunately far from a comprehensive genetic assessment of hereditary risk, and many persons will incorrectly infer that risk is not increased because no mutation was detected. When patients discuss DTC genetic testing, PCPs must consider whether it was appropriate and complete given the patient's personal and family history.

As van der Wouden and colleagues note (4), PCPs need to engage directly about results. Even more important, they need to communicate that they are actively considering genetics and genomics in regular care (4). Publicity generated by Angelina Jolie about hereditary cancer syndromes has raised awareness of the direct role of genetics in care (5). If patients do not believe that PCPs are addressing their risk, they will be tempted to understand that risk through such means as DTC genetic testing, which may be inadequate. Identifying patients at high risk for a predisposition to hereditary cancer is a standard of care for PCPs. Providers must obtain a comprehensive 3-generation family history that is updated annually, provide patients with the required information based on family history that is needed to make educated decisions, document discussions with the patient in the medical record, and refer the patient to a CGP for more detailed evaluation when genetic risk is suspected. Primary care providers may be held liable if they do not identify a patient at risk for a predisposition to inherited cancer that may have led to increased surveillance or risk-reducing surgeries.

Primary care providers are uniquely positioned to identify patients at high risk for inherited predisposition, promote the appropriate referral to CGPs for testing services, and facilitate risk management strategies in line with best practice guidelines. They need to be aware of the limitations of DTC genetic testing and be prepared to discuss the role of genetics and genomics in care. Comprehensive care is best provided with a team approach that includes both PCPs and CGPs.

Suzanne M. Mahon, RN, DNSc, AOCN, APNG
Saint Louis University
Saint Louis, Missouri

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Lyme Disease

TO THE EDITOR: In May, the incidence of vector-borne disease increases markedly as more persons in the United States spend time outdoors. Therefore, the publication of Hu's *In the Clinic* article on Lyme disease (1) in May 2016 was opportune. However, Hu provides little new information to help clinicians with decision making and repeats outdated dogma.

First, Hu claims that ticks need to be attached for 24 hours for *Borrelia burgdorferi* to be transmitted; however, the minimum attachment time for transmission has never been proved (2). As Hu states, many patients with Lyme disease are unaware of having been bitten.

Second, the recommendation to use single-dose doxycycline for prophylaxis after a tick bite contradicts the evidence that supports treatments lasting 20 days or longer (3).

Third, resolution of erythema migrans does not indicate that treatment has been successful. The eruption resolves on its own, increasing clinical confusion about disease duration and the efficacy of treatment.

Fourth, the fundamental challenge in Lyme disease is the lack of a gold standard to exclude infection or to prove cure. Hu states that diagnostic tests for this condition have a high incidence of false-positive results, yet chronic infection with *B burgdorferi* leads to immune depression and hypogammaglobulinemia (4). Indeed, substantial evidence shows high rates of false-negative results on serologic tests for Lyme disease. Patients with this condition and their physicians must recognize that the results of serologic tests alone neither prove nor disprove infection. In contrast, a positive culture for *B burgdorferi* is compelling evidence of active infection. A reliable method for culture is commercially available but not widely used (5).

Fifth, this absence of a gold standard for diagnosis and cure is a fundamental obstacle to progress against Lyme disease. Confused clinical definitions and circular reasoning often equate the disappearance of erythema migrans with the resolution of disease, leading to erratic reporting. For example, Hu coauthored a review published in the *Journal of the American Medical Association* published 1 week before the *In the Clinic* article. In this review, he estimates that 90% of patients with Lyme disease have erythema migrans. In the *In the Clinic* article, his estimate was 70%.

Finally, treatment guidelines that Hu describes contradict those posted on the National Guideline Clearinghouse. Older Centers for Disease Control and Prevention guidelines do not

meet modern standards of proof and are misleading. Patients and clinicians should be referred to the only evidence-based guidelines currently available (3). Reviews that obfuscate and mislead do not help the more than 300 000 persons who will contract Lyme disease in the United States this year or the millions who already have complications of this infection.

Daniel Weiss, MD

Your Diabetes Endocrine Nutrition Group
Mentor, Ohio

Miriam F. Weiss, MD, MA

Case Western Reserve University
Cleveland, Ohio

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IN RESPONSE: First, I agree that no study has shown the transmission time from an infected tick to a human. Estimates for the time of attachment needed to transmit infection are extrapolations from animal studies that may not accurately reflect human transmission. And, as stated, results of studies of the efficacy at preventing disease of removing feeding ticks before 24 to 48 hours have been mixed. However, early removal of attached ticks poses little downside risk.

Second, recommendations for consideration of single-dose doxycycline prophylaxis are based on results of a randomized, controlled trial (1). It is not unusual for recommendations for prophylaxis and treatment of disease to differ.

Third, I agree that resolution of erythema migrans alone is not sufficient to establish treatment success. In most treatment studies, resolution of this rash was not used as the sole criterion for success of therapy. Most studies also compared time to resolution, need for retreatment, and development of late sequelae.

Fourth, as I stated in the article, no currently available tests offer "proof of cure" and serologic testing cannot distinguish between active and treated infection. Although some causes of false-negative results on serologic testing are

known, immune depression and hypogammaglobulinemia are not thought to be related to *B burgdorferi* infection. I do fully endorse the statement that serologic tests alone neither prove nor disprove infection. The culture method cited by Drs. Weiss and Weiss has not been independently validated, and substantial questions have been raised about the results (2).

Fifth, I agree that better definitions and diagnostic tests are needed in the controversial area involving patients with persistent symptoms after treatment of Lyme disease.

Finally, the guidelines posted in the National Guideline Clearinghouse are from the International Lyme and Associated Diseases Society and have not been endorsed by the National Guidelines Clearinghouse or other major medical societies. Although I agree that higher-quality data on the treatment of many aspects of Lyme disease are needed, the presented recommendations are up to date based on currently available information.

Linden T. Hu, MD

Tufts University School of Medicine
Boston, Massachusetts

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OBSERVATIONS

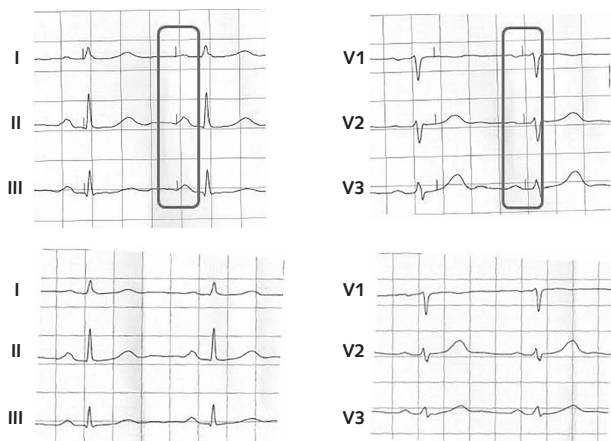
Neuromodulation of Gastric Muscle Can Affect the Electrocardiogram: A Case Report

Background: Neuromodulation in visceral medicine involves surgical insertion of a device, placement of electrodes in a neural plexus or the muscle layer of an organ, and delivery of electrical pulses with specific energies and intervals. It is currently used to manage fecal incontinence (by stimulation of the sacral nerve), symptoms of gastroesophageal reflux (by stimulation of the lower esophageal sphincter), and gastroparesis (by stimulation of the neurons in gastric muscle) (1).

Objective: To alert clinicians to the possibility that neuromodulation devices can complicate the interpretation of electrocardiograms (ECGs).

Case Report: We have 3 patients with severe gastroparesis who have been equipped with the Enterra device (Medtronic) to manage symptoms of nausea by stimulating the neurons in gastric muscle. Gastroparesis was caused by diabetic neuropathy in 2 women aged 27 and 47 years and by prior surgery in a 52-year-old man. Electrocardiography performed routinely in all 3 patients after placement of the devices showed spikes with an amplitude of about 0.4 mV and a

Figure. ECG before deactivation of the gastric electrical stimulation device showing a spike in ECG leads (0.4 mV and 100 beats/min) (top panels) that disappeared after deactivation (bottom panels).



ECG = electrocardiogram.

frequency of 100 beats/min in all leads (Figure), which reflects the stimulation pattern of the device (0.1 second “on” and 0.5 second “off”). We ensured correct function of the neuromodulation devices through clinical examination and device interrogation, and we ruled out lead dislodgement with abdominal radiographs. None of the patients had cardiac symptoms, a known cardiac disorder, or a cardiac implantable device. Electrocardiograms from before implantation of the neuromodulation devices showed no abnormal spikes and were otherwise unremarkable. Deactivation of each neuromodulation device resulted in immediate disappearance of the abnormal spikes.

Discussion: Although the pathophysiologic mechanisms of the effects of neuromodulation are not well-understood, it is being used more frequently to treat functional gastroenterologic disorders; thus, the effects we describe are likely to occur more frequently (2). All neuromodulation devices stimulate afferent and efferent neurons with high-frequency and low-voltage electric current. The devices used in patients with gastroparesis to stimulate gastric neurons for the management of nausea typically use a frequency of 14 to 30 Hz and a stimulating current of 5 to 10 mA (1), and they pulse with a distinct pattern (0.1 second on and 0.5 second off). To our knowledge, there is only 1 other report of a visceral neurostimulator causing ECG findings (3). That report described ECG spikes similar to the ones described here, but in a patient equipped with a device stimulating the sacral nerve to manage fecal incontinence. In that case, a lead was dislodged, and the spikes disappeared after the lead was repositioned. We also are aware of a single report of a patient who was equipped with an implantable cardiac defibrillator and a device to stimulate gastric muscle (4). Interrogation studies showed no interactions between the devices.

We are convinced that the ECG spikes we observed in our patients came from their neurostimulation devices. Therefore, we advise other clinicians to consider obtaining baseline

ECGs before implanting similar devices to aid in the interpretation of ECGs obtained after device implantation.

Maximilian Tiller, MD
Sebastian Reif, MD
Stefan Volz, MD
Wolfgang Schepp, MD
Felix Gundling, MD
Bogenhausen Academic Teaching Hospital,
Munich Municipal Hospital Trust
Munich, Germany

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Injuries From an Exploding E-Cigarette: A Case Report

Background: E-cigarettes simulate tobacco smoking and are a popular alternative to traditional cigarettes.

Purpose: To report what we believe to be the first case in the medical literature of an e-cigarette explosion involving injury to the user.

Case Report: A 23-year-old man was transferred to our hospital for management of injuries sustained when an e-cigarette exploded in his mouth, resulting in injuries and burns to his face, left hand, and chest. Before being transferred, he was stabilized and intubated for airway protection because of the possibility of inhalation injury. Computed tomography revealed comminuted fractures of his hard palate and nasal septum, with dislocation of adjacent teeth (Figure). Digital radiography of his left hand showed an intra-articular fracture of his left fifth proximal phalanx, and ophthalmologic examination revealed a large corneal abrasion. He was admitted to the trauma burn intensive care unit for 4 days, and multiple bronchoscopies revealed inhalation injuries. Due to his injuries, including a defect noted in the soft palate on physical examination, clinical concern was raised for a possible oropharyngeal fistula. However, no reflux of liquids into the nasopharynx was noted after his diet was advanced, so the decision was made for conservative management of the maxillofacial injuries during the inpatient stay. He was transferred to the stepdown unit for pain control, and after 8 days in the hospital, was discharged on a regimen of oral antibiotics and pain medication. The patient was scheduled to follow up in 1 month for surgical repair of his hard palate and was advised

not to have dental implants made for the missing teeth until after his palate was repaired.

Discussion: More than 460 brands of e-cigarettes are available, and many brands have multiple configurations. The basic components are mouthpiece, liquid reservoir, heating element, and power source. The power source can be either an external source or a battery that can be recharged or replaced. The presumed cause of spontaneous e-cigarette explosions and fires is lithium-ion batteries, which have caused fires and explosions in cell phones, laptop computers, medical devices, and electric cars (1, 2). These batteries can experience thermal runaway, whereby increases in temperature cause an uncontrolled exothermic reaction among the anode, cathode, and flammable electrolytes within the battery. This results in increasing temperature and pressure, which can cause the battery to rupture, leading to a fire, an explosion, or both (2). Most e-cigarettes are cylindrical, and the battery is at the end near the mouthpiece, which is the weakest structural point of the device (1). If a battery were to fail while in the user's mouth, it would cause the type of injuries observed in our patient.

We believe that further research on this problem is desirable because there are no governmental regulations in the United States with regard to the safety of e-cigarette batteries and because these cigarettes have increased in popularity since their release in 2007. For example, between 2009 and 2011, the percentage of the U.S. population reporting e-cigarette use increased from 0.6% to 6.1% (3). In addition, the Centers for Disease Control and Prevention reported that the number of middle school and high school students who experimented with e-cigarettes nearly doubled between 2011 and 2012 (4). Also, there were an estimated 2.5 million e-cigarette users in the United States in 2014 (5).

Daniel E. Cason, MD
Desiree E. Morgan, MD
Jason A. Pietryga, MD
UAB Department of Radiology
Birmingham, Alabama

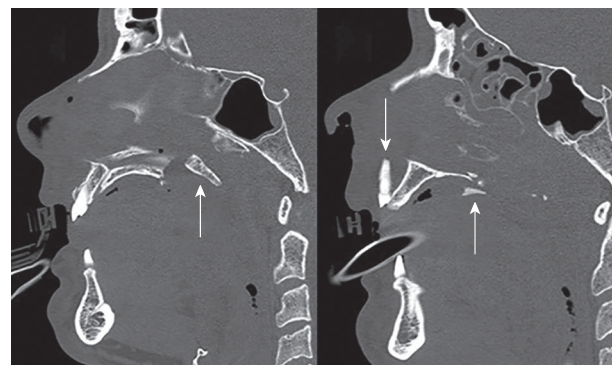
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Figure. CT images of the patient's injuries.



CT = computed tomography. **Left.** Sagittal reformatted maxillofacial CT showing a comminuted fracture of the hard palate, with posterior displacement resulting in an oroantral fistula (arrow). **Right.** Sagittal reformatted maxillofacial CT showing anterior and superior dislocation of the ninth tooth (downward arrow) and comminuted fracture of the hard palate (upward arrow).

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CORRECTION

The guidelines referenced in a recent editorial (1) were from the American College of Gastroenterology, not the American Gastroenterological Association. Also, more individuals were febrile with autologous stool, not with donor stool.

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Reference

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